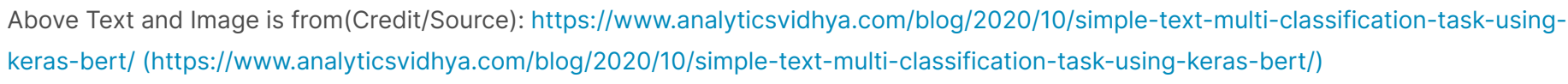


3 Copy & Edit 22

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- Token embeddings: A [CLS] token is added to the input word tokens at the beginning of the first sentence and a [SEP] token is inserted at the end of each sentence.
- Segment embeddings: A marker indicating Sentence A or Sentence B is added to each token. This allows the encoder to distinguish between sentences.
- Positional embeddings: A positional embedding is added to each token to indicate its position in the sentence.



```
In [1]: import warnings
warnings.filterwarnings("ignore")

In [2]: from transformers import * # this is HuggingFace library
import tensorflow as tf
from transformers import BertTokenizer, TFBertModel, BertConfig
```

Pretrained model on English language using a masked language modeling (MLM) objective. It was introduced in this paper and first released in this repository. This model is uncased: it does not make a difference between english and English.

Credit/Source: <https://huggingface.co/bert-base-uncased> (<https://huggingface.co/bert-base-uncased>)

1. BERT-Base, Uncased: 12-layer, 768-hidden, 12-heads, 110M parameters
2. BERT-Large, Uncased: 24-layer, 1024-hidden, 16-heads, 340M parameters
3. BERT-Base, Cased: 12-layer, 768-hidden, 12-heads, 110M parameters
4. BERT-Large, Cased: 24-layer, 1024-hidden, 16-heads, 340M parameters
5. BERT-Base, Multilingual Case: 104 languages, 12-layer, 768-hidden, 12-heads, 110M parameters
6. BERT-Base, Chinese: Chinese Simplified and Traditional, 12-layer, 768-hidden, 12-heads, 110M parameters

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Downloading: 100% 466k/466k [00:00<00:00, 1.14MB/s]

Downloading: 100% 570/570 [00:00<00:00, 21.3kB/s]

```
Downloading: 100% 536M/536M [00:12<00:00, 45.3MB/s]
```

Below is example how BERT Uncased Tokenizer works:

- Credit/Source: <https://swatimeena989.medium.com/bert-text-classification-using-keras-903671e0207d#b60a>
(<https://swatimeena989.medium.com/bert-text-classification-using-keras-903671e0207d#b60a>)

You can see that Uppercase words are also converted into smallcase.